THE SOFTWARE DEVELOPMENT LIFE CYCLE

SDLC

OVERVIEW

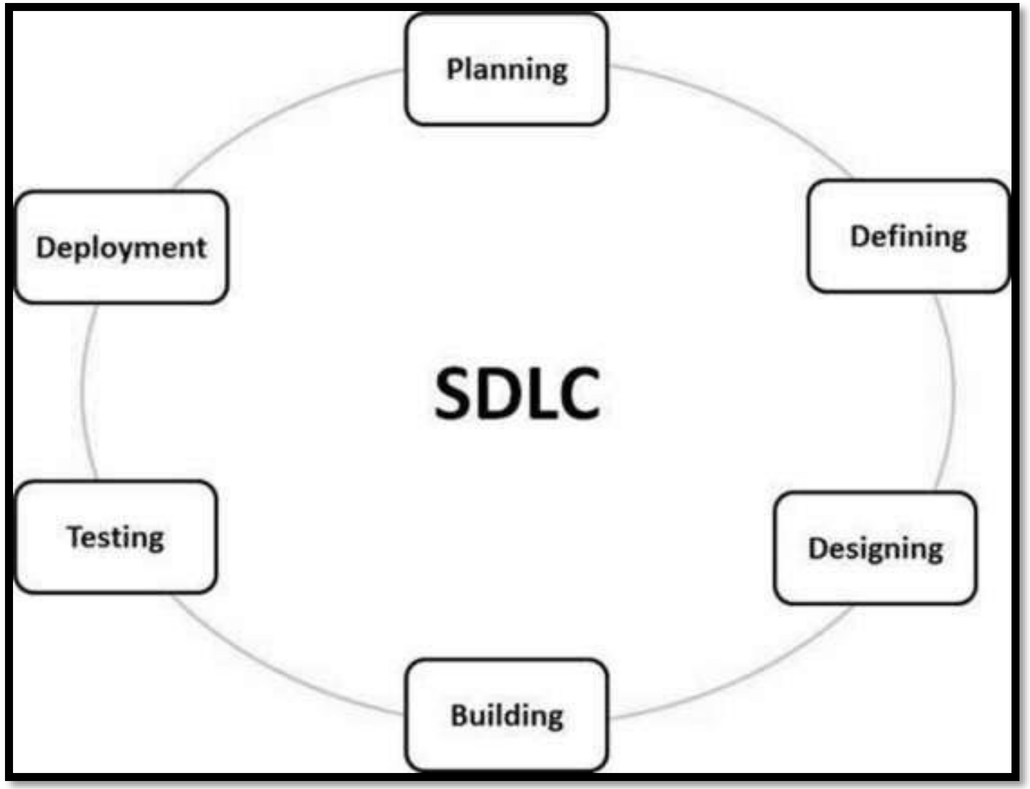
- What is SDLC?
- SDLC Stages
- SDLC Methodologies
  - Big Bang
  - Waterfall
  - Agile
    - Scrum framework
    - Kanban framework

WHAT IS SDLC?

- SDLC stands for Software Development Life Cycle
- When building any type of application, there is a method dev teams follow for organizing, managing, and building the application
- These methods of development are known as the software development life cycle

SDLC STAGES

- A typical SDLC consists of these stages:
  - Planning – takes input from customer and other experts to plan the project approach
  - Defining/Analysis - clearly define and document requirements and have the customer approve them
  - Designing – create software architecture, prototype, and user experience design
  - Building/Implementing - developers work on actually writing the code
  - Testing – focuses on testing the written code for bugs and inconsistencies
  - Deployment – launching the fully-fledged product on the market
  - Maintenance – adding new functionality and updating with bug fixes



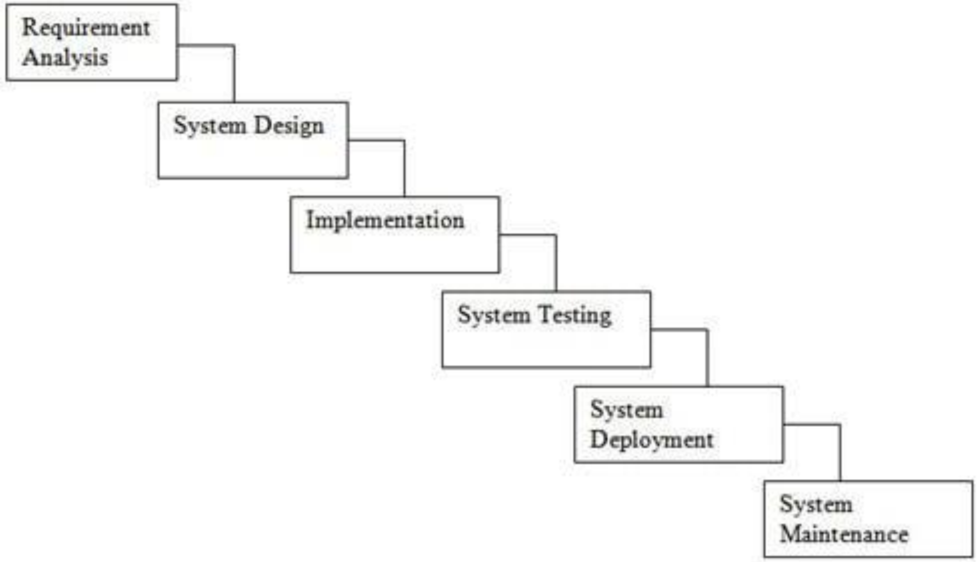
BIG BANG METHODOLOGY

- The Big Bang methodology is an SDLC model that does not follow any specific process
- There is no defined procedure for the Big Bang Model
- There is very little planning
- The client/customer is also not sure what they want
- Requirements are implemented on the fly
- This model is usually for small projects where development teams are very small

WATERFALL METHODOLOGY

- The Waterfall model was the first SDLC model to be widely used
- It is also known as a linear-sequential life cycle model
- Each phase must be completed before the next phase can begin.
- There is no overlapping phases and no returning to a previous phase

WATERFALL METHODOLOGY



Waterfall Model - © www.SoftwareTestingHelp.com

BENEFITS OF WATERFALL

- Simple and easy to understand and use
- Model is rigid, making it easy to manage
- Phases are processed and completed one at a time
- Clearly defined stages
- Process and results are well documented
- Works well for smaller projects where requirements are very well understood

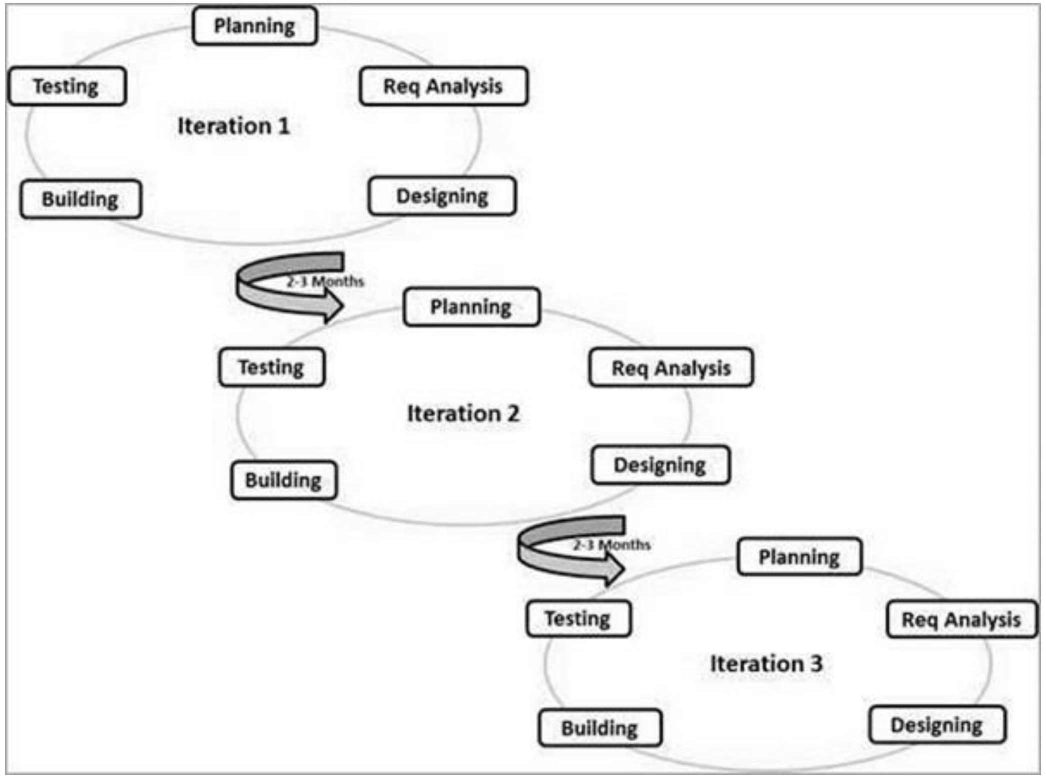
DISADVANTAGES OF WATERFALL

- No working software is produced until late in the life cycle
- Poor model for long and ongoing projects
- It is difficult to measure progress within stages
- Cannot accommodate changing requirements
- High amounts of risk and uncertainty

AGILE METHODOLOGY

- The Agile model is a combination of iterative and incremental process models
- Focuses on adaptability and customer satisfaction by rapid delivery of working software product
- Each iteration involves cross-functional teams working simultaneously on all phases of the SDLC
- After each iteration, a working product is displayed to the customer
- Agile development teams divide project iterations into **Sprints** and produce working software product at the end of each sprint

AGILE METHODOLOGY



AGILE SPRINTS

- A sprint typically lasts two weeks or 10 business days
- Each sprint typically contains the following SDLC stages:
  - Planning
  - Defining/analysis
  - Development
  - Testing
  - Assessment (by the customer)
  - Delivery

AGILE STORIES

- In agile, all project requirements are known as user 'stories'
- A story is the smallest unit of work in an agile framework
- It is an end goal expressed from the software user's perspective
- Stories are informal, general explanations of a software feature
- Ex: 'as a user, I want to be able to transfer money between accounts'
- The combination of all stories make up the entire project known as the 'epic'
- Tasks are the actual work of implementing a story

SCRUM FRAMEWORK

- The scrum framework defines a set of meetings, tools, and roles help with team management
- Scrum involves a **product owner** that maintains a **backlog** of all user stories
- A scrum master is a team member tasked with holding the team accountable to the agile ceremonies
  - Ex: a scrum master hosts daily standup meetings and leads retrospectives

SCRUM CEREMONIES

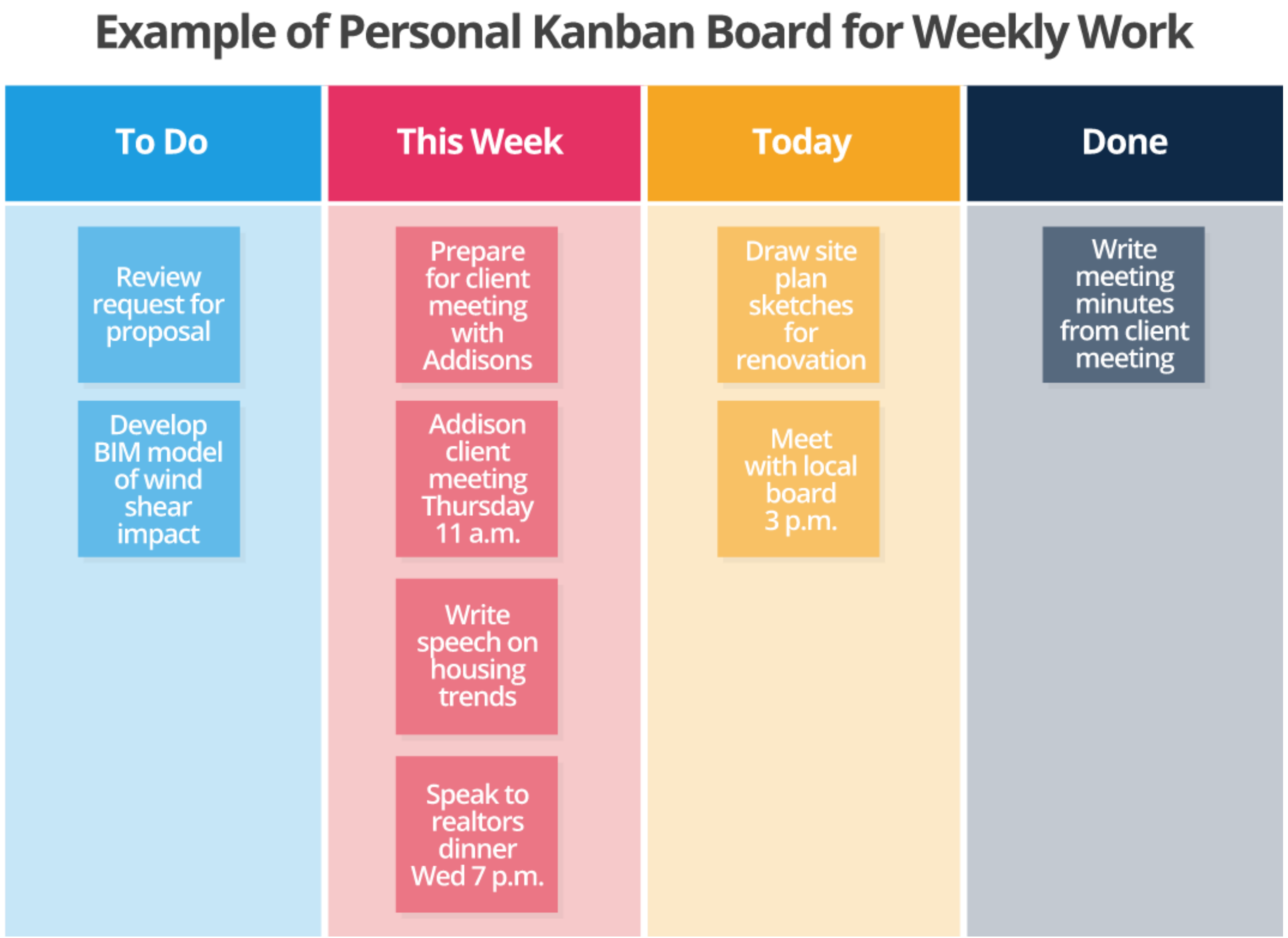
- Scrum ceremonies are meetings with defined lengths, frequencies, and goals
- They are meant to help project teams plan, track, and engage stakeholders
- Key ceremonies include:
  - Sprint planning ceremony – determine the goals, participants, and length of the sprint
  - Daily Standup Ceremony – lets the team discuss their progress, any problems, and their goals for the day
  - Sprint Review - discuss and demo work completed, highlight accomplishments
  - Sprint Retrospective – understand what worked last sprint, what didn't work, and how to improve on the next sprint

SCRUM FRAMEWORK

KANBAN FRAMEWORK

- Kanban is an agile framework that makes use of a Kanban board
- It is a method for defining, managing, and improving work amongst teams
- A Kanban board is where teams can visually display project progress
- The product backlog, stories in progress, and stories that meet the definition of 'done' are all displayed on the Kanban board

KANBAN BOARD



ADVANTAGES OF AGILE

- Is a very realistic approach to software development
- Promotes teamwork and cross training
- Functionality can be developed quickly and demonstrated
- Good model for changing or fixed requirements
- Little planning needed
- Easy to manage
- Flexible for developers
- Puts customer needs first

DISADVANTAGES OF AGILE

- Not suitable for handling complex dependencies
- There is a very high individual dependency, since there is minimum documentation generated.
- Limited documentation
- No finite end to the project
- Difficult to determine completion date as sprints are 'see as you go'